

Roy S Hung

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OBJECTIVE

Automation and Systems Engineer with 6+ years of experience developing software, automation, and data-driven solutions for engineering organizations. Holds a Master of Applied Science in Computer Science from the University of Pennsylvania (GPA 3.92/4.0). Skilled in full-stack development, backend systems, database optimization, and machine learning applications, with a proven track record of delivering solutions that improve operational efficiency and decision-making.

EDUCATION

University of Pennsylvania GPA 3.92/4.0 January 2024 – April 2026

Master of Applied Science, Computer Science (Formerly MCIT)

Coursework: Data Structures, Database, Algorithms & Computation, Artificial Intelligence, Machine Learning in Data Science

The University of Texas at Austin

Bachelor of Science in Chemical Engineering *Honors*

SKILLS

- **Programming Languages:** Python, Java, C, JavaScript/TypeScript, HTML/CSS
- **Frameworks:** React, Node.js, Express.js, JUnit, Tailwind CSS
- **Databases:** PostgreSQL, MongoDB (NoSQL)
- **Tools & Platforms:** GitHub, Docker, AWS, Valgrind, GDB Debugger, Visual Studio, DataGrip, Streamlit, SpotFire
- **Core Competences:** Natural Language Processing (NLP), Statistical Modeling, Full-Stack Web Application Development, RESTful API & Backend System Design, Cloud Deployment (AWS), Database Optimization, Workflow Optimization

PROJECTS

Relocation Search Engine | ReactJS, Node.js, Express, PostgreSQL, MongoDB, Tailwind, Pandas, AWS July 2025

- Collaborated in a four-member team to build a full stack web application with PostgreSQL relational database to query and visualize population, housing prices, and demographic data at ZIP/City/State level
- Engineered preprocessing scripts for over 10 data sources and integrated them into a unified PostgreSQL schema, enabling efficient query performance for downstream web application development
- Optimized database performance by implementing materialized views and indexing to reduce API response time from 15 seconds to under 1 second
- Implemented secure user authentication using Firebase OAuth (Google/GitHub) with MongoDB for scalable session storage

Wikipedia Biography Parser | Python, OpenAI API August 2025

- Optimized a generative AI model with GPT-4.1 to output coherent biographies from structured Wikipedia Infobox data
- Built a reverse information extraction system to parse unstructured text and convert it into structured attribute fields

EXPERIENCES

Samsung Electronics – Senior Automation & System Engineer June 2026 – Current

- Developed innovation automation solutions for customer automation needs
- Maintained complex systems with continual functional and documentation improvements

University Of Pennsylvania – Computer System Programming Teaching Assistant Jan 2026 – Current

- Lead recitation sessions for 200+ students covering memory management, processes, and UNIX system calls in C
- Mentor students on debugging, segmentation faults, and low-level program behavior using GDB and Valgrind

Samsung Austin Semiconductor – Dry Etch Process Improvement Engineer II March 2020 – November 2023

- Developed and deployed a Python/Streamlit web application to analyze historical process data and apply statistical models to predict optimal run times across 50+ product lines and improving engineering input accuracy.
- Automated process gas adjustment calculation with Python dashboard (Pandas, Streamlit) to save \$20,000 annually
- Engineered a quality incident investigation app using SQL, Python, and Streamlit to query and organize process time data and historical equipment health trends, reducing root cause investigation time by 5 hours per incident

Samsung Austin Semiconductor – Process Operation Engineer I January 2018 – March 2020

- Developed automated SPC summary report tool with SQL/VBA to eliminate manual reviews of over 100 control parameters
- Built automation tools for data aggregation and visualization in VBA to streamline equipment qualification analysis, reducing reporting time from 30 minutes to 5 minutes